

Lightweight Multimodal Person Detection and Temporal Post-Processing for Aerial Search and Rescue

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Abstract—Autonomous unmanned aerial vehicles (UAVs) in wilderness search and rescue (SAR) require reliable person detection under strict size, weight, power, and computing (SWaP-C) constraints. While multimodal sensing combining visible red-green-blue (RGB) and thermal infrared (TIR) cameras mitigates single-modality failures from canopy shade, darkness, or thermal crossover, frame-level detectors frequently produce transient false positive spikes. In autonomous flight, each false alarm causes wasteful loiter maneuvers, battery drain, and spurious transmissions over bandwidth-limited Internet of Things (IoT) satellite or cellular links. This paper evaluates lightweight causal temporal decision methods for multimodal aerial SAR alert generation. Using a controlled 2×2 experimental design across two biomes (arid desert and temperate forest) and two modalities (RGB and thermal), we deploy two upstream You Only Look Once (YOLO11n) detectors (2.6M parameters, 6.5 GFLOPs at 640×640) trained with batch size 16 in single-precision floating-point (FP32) arithmetic. Detection confidences are integrated via soft disjunctive late fusion and filtered across a causal five-frame sliding window (208 ms at 24 Hz). We benchmark five temporal decision strategies under validation-optimized thresholds: baseline raw thresholding, five-frame moving average, five-frame median filtering, five-frame history consensus, and a learned pure-PyTorch Mamba selective state-space model (SSSM). The methods are evaluated across positive event detection, false alert suppression, decision latency, and computational overhead.

Index Terms—Multimodal fusion, search and rescue, unmanned aerial vehicles, state-space models, lightweight object detection.

I. INTRODUCTION

Autonomous unmanned aerial vehicles (UAVs) are critical tools in wilderness search-and-rescue (SAR) operations [1], [2]. In remote terrains such as arid deserts and dense forests, aerial robotic platforms survey disaster sectors far faster than ground search crews. However, onboard computer vision systems operate under severe size, weight, power, and computing (SWaP-C) constraints. Furthermore, remote SAR missions lack broadband communication infrastructure. UAVs must transmit

alarm telemetry over low-bandwidth cellular or satellite Internet of Things (IoT) uplinks [3]. Under these conditions, false alarms carry severe operational penalties: each unverified alert diverts the drone to execute a hover-and-verify loiter maneuver, depleting battery reserves, or dispatches rescue teams across hazardous terrain.

Wilderness optical sensing exhibits modality-specific failure modes. Visible red-green-blue (RGB) cameras degrade in dense canopy shadows, fog, and darkness. Conversely, thermal infrared (TIR) cameras detect emitted human body heat regardless of illumination [4], [5]. However, thermal sensors encounter thermal crossover when ground temperatures equilibrate with human skin radiation [6]. Sun-warmed boulders, heated sand, and wildlife produce thermal anomalies that trigger false detections. Reliable aerial search therefore requires dual-spectrum sensing where visible and thermal streams provide complementary evidence.

To satisfy real-time constraints on edge processors, aerial platforms rely on compact single-stage detectors such as You Only Look Once (YOLO) [7], [8]. In this work, we deploy Ultralytics YOLO11n [9] (2.6 million parameters, 6.5 giga-floating-point operations (GFLOPs) at 640×640 resolution). To prevent half-precision numerical instability on embedded graphics processing units (GPUs), we train the detector with batch size 16 in single-precision floating-point (FP32) arithmetic. Although lightweight models achieve high inference throughput, frame-level detections exhibit temporal volatility. Confidence scores oscillate from platform vibration, foliage occlusions, viewpoint changes, and fleeting distractors. Direct thresholding of raw detections causes isolated confidence spikes to trigger false alarms on empty scenes and transient dips to miss valid targets.

To resolve these challenges without conflating model architecture with domain generalization, we freeze our study around a controlled 2×2 experimental design across two environments (arid desert and temperate forest) and two visual modalities (visible RGB and thermal infrared). We train two separate upstream YOLO11n detectors: one specialized

This work involved human subjects or animals in its research. The author(s) confirm(s) that all human/animal subject research procedures and protocols are exempt from review board approval.

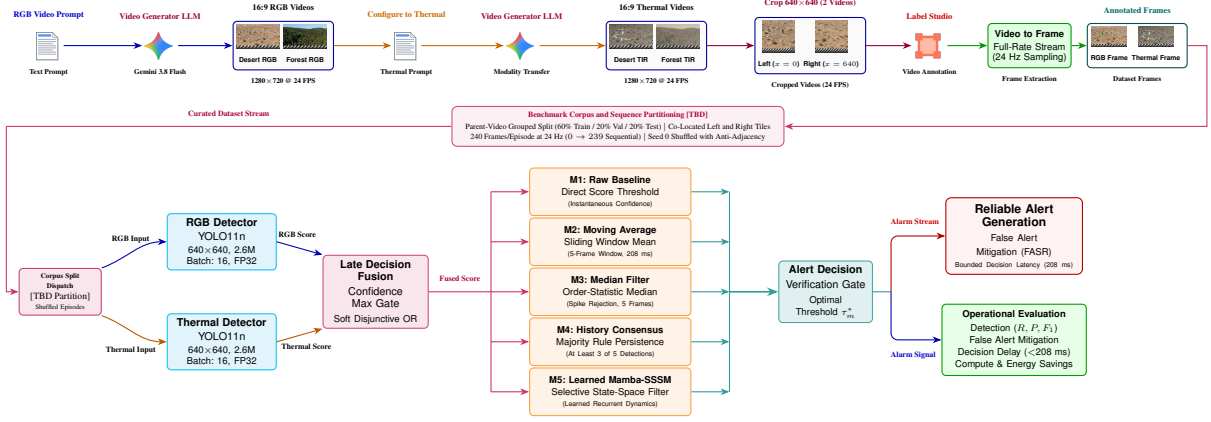


Fig. 1. System architecture of the MMSAR evaluation framework. Visible RGB and thermal infrared video streams are synthesized from text prompts, followed by cross-modality thermal transfer, dual-crop (640×640) spatial extraction, video-level annotation in Label Studio, and 24 Hz frame extraction. Curated sequences are partitioned into 240-frame episodes under a parent-video grouped split protocol ([TBD]) with anti-adjacency snippet shuffling to eliminate cross-tile spatial leakage. At edge inference, parallel lightweight YOLO11n detectors process incoming RGB and thermal frames at 24 Hz ($T_s \approx 41.7$ ms), integrated via soft disjunctive late fusion ($s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$) and evaluated across five lightweight temporal decision methods ($W = 5$ frames, 208 ms latency bound).

on multi-environment RGB data and one on thermal data. Downstream, causal temporal post-processing filters the fused confidence stream across a causal sliding window of five frames (208 ms at 24 Hz). We evaluate deterministic smoothing filters, order-statistic filters, temporal consensus tracking, and a learned Mamba selective state-space model (SSSM) [10], with each method operating under an optimal decision threshold calibrated on validation sequences.

The central research question is: *How do lightweight causal temporal decision methods compare in generating reliable and efficient alerts from multimodal RGB and thermal detections for autonomous aerial search and rescue?*

This work delivers five main contributions: (1) **Controlled 2×2 Video Corpus**: A dual-stream RGB and thermal benchmark across arid desert and temperate forest biomes, structured into positive targets, hard-negative distractors (heated rocks, moving shadows), and clear-negative terrain. (2) **Modality-Specific Detection Baselines**: Standardized edge detectors deploying two separate YOLO11n models (2.6M parameters, 6.5 GFLOPs at 640×640) trained on multi-environment RGB and thermal sequences in FP32 arithmetic. (3) **Multimodal Alert Decision Benchmark**: An empirical evaluation of five lightweight temporal decision techniques (raw thresholding, moving average, median filtering, history consensus, and Mamba-SSSM) across event detection, false alarm suppression, decision delay, and processing overhead under validation-optimized thresholds. (4) **Modality and Environment Sensitivity Analysis**: Systematic characterization of temporal decision filters across RGB and thermal confidence streams and disparate wilderness biomes. (5) **Operational Resource Impact Analysis**: Formal modeling of downstream mission benefits in robotic battery life and IoT telemetry bandwidth conservation.

The remainder of this paper is structured as follows: Section II reviews related literature; Section III details the

multimodal search-and-rescue (MMSAR) sensing, fusion, and causal post-processing methodology; Section IV outlines the experimental setup; Section V evaluates detection benchmarks, operational impacts, and experimental limitations (Section V-D); and Section VI concludes the paper.

II. RELATED WORK

A. Multispectral and Multimodal Person Detection

Multispectral sensing combines complementary spectral bands to overcome illumination extremes. Hwang et al. [4] introduced the KAIST benchmark, proving that thermal radiation preserves human body signatures when visible lighting fails. Jia et al. [5] released LLVIP for visible-infrared surveillance. Regarding fusion architecture, Liu et al. [11] demonstrated that halfway and late fusion preserve modality-specific features better than early concatenation. Li et al. [12] developed illumination-aware weighting, while Zhang et al. [13] used cross-modality attention. In wilderness SAR, canopy shading and solar heating vary dynamically. We adopt soft disjunctive late fusion via max-pooling, allowing each detector branch to execute independently without cross-channel synchronization bottlenecks.

B. Lightweight Detectors for Aerial Search and Rescue

Deploying vision models on UAVs requires balancing detection accuracy against SWaP-C constraints. Rudol and Doherty [1] demonstrated dual-band person geolocation for aerial search, while Silvagni et al. [2] stressed that battery capacity bounds mission endurance. Lygouras et al. [14] showed that heavy neural ensembles exceed embedded compute envelopes. Portmann et al. [6] identified thermal crossover as a primary source of aerial false alarms. Kyrkou and Theodorides [3] addressed telemetry bottlenecks with EmergencyNet, showing that spurious alerts congest low-bandwidth uplinks. Single-stage object detectors have progressed from YOLO [7] through

TABLE I
SUMMARY OF KEY MATHEMATICAL NOTATIONS AND SYSTEM
PARAMETERS

Symbol / Variable	Physical Description / Operational Definition
$I_{\text{RGB}}[n], I_{\text{Thermal}}[n]$	Synchronized visible RGB and thermal infrared video frames at index n
$c_{\text{RGB}}[n], c_{\text{Thermal}}[n]$	Upstream YOLO11n detection confidence scores $\in [0.0, 1.0]$
$s[n]$	Fused confidence score via soft disjunctive late fusion $\in [0.0, 1.0]$
f_s, T_s	Operating timebase ($f_s = 24$ Hz, interval $T_s \approx 41.7$ ms)
$W, \mathcal{W}[n]$	Sliding window length ($W = 5$ frames, 208 ms) and observation vector
$\mathcal{M}_m, \tau_m^*$	Causal filter method $m \in \{M1, \dots, M5\}$ and optimal validation threshold
$\mathcal{D}_{\text{val}}, \Delta\tau$	Validation sequence partition and sweep step size ($\Delta\tau = 0.02$)
$y[n]$	Binary alarm decision asserted at frame n ($y[n] \in \{0, 1\}$)
$d_{\text{model}}, d_{\text{state}}$	Hidden projection dimension (16) and recurrent state dimension (8)
A, B_t, C_t, Δ_t	SSSM state transition, input, output, and selective discretization parameters
h_t, u_t	Recurrent hidden state tensor and selective projection representation
$S_{\text{pkt}}, K_{\text{FP}}$	Telemetry packet size (1.2 kB standard, 45 kB thumb) and false alarm count
Ω_{wasted}	Cumulative wasted uplink telemetry bandwidth
$P_{\text{hover}}, T_{\text{loiter}}$	UAV hovering power draw (280 W) and loiter duration (15–30 s)
$E_{\text{loiter}}, \text{FASR}$	Loiter battery energy expenditure and False Alarm Suppression Ratio
$T_{\text{RGB}}, T_{\text{Thermal}}$	Inference latency per frame of YOLO11n-RGB and YOLO11n-Thermal
$T_{\text{fusion}}, T_{\text{postproc}}$	Latency per frame of decision fusion gate and causal post-processing filter

YOLOv7 [8], YOLOv9 [15], YOLOv10 [16], and YOLO11 [9]. We deploy Ultralytics YOLO11n (2.6M parameters, 6.5 GFLOPs) to maximize edge inference efficiency.

C. Temporal Post-Processing and Sequential Decision Making

Per-frame detectors applied to video produce high-frequency confidence chatter. In video detection, Seq-NMS [17] and T-CNN [18] construct temporal box tubelets. However, multi-frame spatial tracking introduces optical flow and graph optimization overhead unsuitable for edge UAVs. Operating directly on scalar confidence streams avoids spatial tracking latency. Foundational sequential detection traces to Wald’s sequential probability ratio test [19]. In radar signal processing, Richards [20] formulated sliding-window M -out-of- N binary integration to reject transient pulses while bounding latency. Rabiner et al. [21] established the noise-rejection properties of running median and moving average filters. We adapt these principles to causal five-frame confidence streams under validation-optimized decision thresholds.

D. State-Space Sequence Models

Recurrent neural networks suffer from vanishing gradients over long sequences, whereas Transformers scale quadratically with sequence length. Structured state-space models (SSMs) provide linear sequence scaling with recurrent execution. Gu et al. [22] introduced S4 to model continuous-time state spaces efficiently. Gu and Dao [10] proposed Mamba, introducing data-dependent selection mechanisms with hardware-efficient parallel scans. Vision adaptations include Vision Mamba [23] and VMamba [24], while Dao and Gu [25] established structured state-space duality connecting recurrence and attention. In this work, we adapt a pure-PyTorch selective SSM (Mamba-SSSM) as a causal temporal post-processor to filter environmental noise spikes.

III. METHODOLOGY

The proposed multimodal search-and-rescue (MMSAR) framework provides an end-to-end pipeline combining lightweight dual-modality detection, soft disjunctive late fusion, and causal temporal post-processing (Fig. 1). Table I outlines key mathematical notations and system parameters.

A. Video Stream Ingestion and Upstream Modality Detectors

Embedded companion computers (e.g., NVIDIA Jetson Orin Nano, 7–15 W thermal design power (TDP)) on autonomous UAVs require real-time processing matching sensor input rates. Synchronized video streams are ingested and processed at $f_s = 24$ Hz ($T_s \approx 41.7$ ms), providing continuous 24 FPS monitoring over harvested 640×640 snippets.

To guarantee causal, frame-synchronous execution without buffering delays or frame drops, the pipeline per frame must satisfy the real-time budget for serial execution (Eq. (1)) or parallel multi-threaded branch execution (Eq. (2)):

$$T_{\text{RGB}} + T_{\text{Thermal}} + T_{\text{fusion}} + T_{\text{postproc}} \leq T_s \approx 41.7 \text{ ms}, \quad (1)$$

$$\max(T_{\text{RGB}}, T_{\text{Thermal}}) + T_{\text{fusion}} + T_{\text{postproc}} \leq T_s \approx 41.7 \text{ ms}, \quad (2)$$

where T_{RGB} and T_{Thermal} represent the inference latencies of the YOLO11n backbones, T_{fusion} is scalar max-pooling latency, and T_{postproc} denotes filter execution latency. Both T_{fusion} ($O(1)$) and T_{postproc} (sub-millisecond across M1–M5, Table II) are computationally negligible compared to deep feature extraction. Consequently, real-time feasibility depends on upstream backbone execution ($T_{\text{RGB}}, T_{\text{Thermal}}$: [TBD - pending on-device benchmarking]).

To isolate temporal post-processing from multimodal co-training artifacts, we train two distinct upstream YOLO11n models [9] (2.6M parameters, 6.5 GFLOPs at 640×640): (1) **YOLO11n-RGB**, trained on multi-environment RGB sequences and yielding confidence $c_{\text{RGB}}[n] \in [0.0, 1.0]$; and (2) **YOLO11n-Thermal**, trained on thermal sequences and yielding confidence $c_{\text{Thermal}}[n] \in [0.0, 1.0]$. If no candidate exceeds non-maximum suppression (NMS), the frame confidence is set to 0.0.

B. Soft Disjunctive Late Fusion Gate

To prevent single-modality dropouts from causing target misses, we integrate per-frame detector outputs via soft disjunctive late fusion using max-pooling:

$$s[n] = \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n]) \in [0.0, 1.0]. \quad (3)$$

The disjunctive formulation ensures that an unambiguous detection in either sensor branch propagates forward with full confidence. Furthermore, decision-level late fusion avoids expensive feature alignment layers, allowing each sensor branch to execute asynchronously on multicore edge processors. Algorithm 1 outlines the per-frame causal alarm pipeline.

C. Causal Temporal Post-Processing

Direct thresholding of raw confidence streams yields high false-alarm rates due to isolated spikes. Downstream processing filters the confidence stream across a causal sliding window of length $W = 5$ frames (208 ms at $f_s = 24$ Hz):

$$\mathcal{W}[n] = \{s[n-4], s[n-3], s[n-2], s[n-1], s[n]\}. \quad (4)$$

Because mathematical filters possess distinct transfer functions, enforcing a uniform threshold degrades performance.

Algorithm 1 MMSAR Causal Alarm Decision (per frame n)**Require:** Synchronized RGB frame $I_{\text{RGB}}[n]$, thermal frame $I_{\text{Thermal}}[n]$ **Ensure:** Binary alarm $y[n] \in \{0, 1\}$

- 1: $c_{\text{RGB}}[n] \leftarrow \text{YOLO11n-RGB}(I_{\text{RGB}}[n])$
- 2: $c_{\text{Thermal}}[n] \leftarrow \text{YOLO11n-Thermal}(I_{\text{Thermal}}[n])$
- 3: $s[n] \leftarrow \max(c_{\text{RGB}}[n], c_{\text{Thermal}}[n])$ {Eq. (3)}
- 4: Update causal window $\mathcal{W}[n] \leftarrow \{s[n-4], \dots, s[n-1], s[n]\}$
- 5: $y[n] \leftarrow \mathcal{M}_m(\mathcal{W}[n]; \tau_m^*)$ {Method $m \in \{\text{M1}, \dots, \text{M5}\}$, Eqs. (6)–(12)}
- 6: **if** $y[n] = 1$ **then**
- 7: Emit telemetry packet (S_{pkt}) and trigger UAV loiter maneuver
- 8: **end if**
- 9: **return** $y[n]$

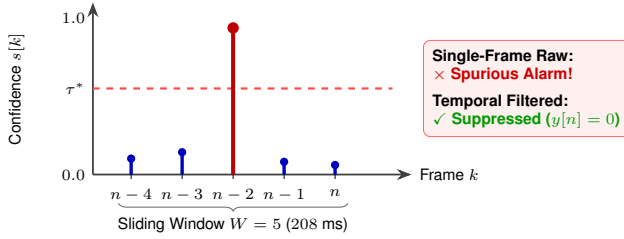
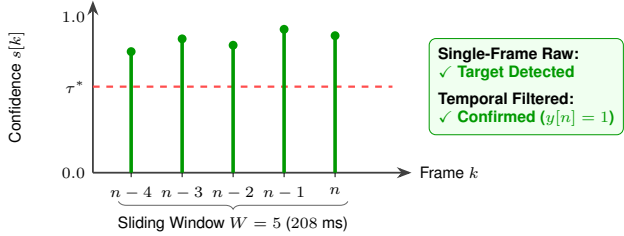
(a) Transient Noise Spike (Single-Frame Clutter)**(b) Sustained Target Signature (True SAR Victim)**

Fig. 2. Temporal signal behavior ($W = 5$ frames ≈ 208 ms at 24 Hz): (a) an isolated noise spike on a hard-negative scene triggers a spurious raw detection but is rejected by causal filtering ($y[n] = 0$), whereas (b) a sustained victim signature confirms target presence and asserts the alarm ($y[n] = 1$).

Each method $m \in \{\text{M1}, \dots, \text{M5}\}$ is assigned an optimal decision threshold τ_m^* calibrated on the disjoint validation set $\mathcal{D}_{\text{val}}$:

$$\tau_m^* = \arg \max_{\tau \in [0.05, 0.95]} F_1(\tau; \mathcal{D}_{\text{val}}, \mathcal{M}_m), \quad (5)$$

sweeping τ with step $\Delta\tau = 0.02$ across 46 candidate thresholds. Once calibrated on $\mathcal{D}_{\text{val}}$, thresholds remain frozen for test evaluation. Complete sweep trajectories are retained to construct operating curves (Fig. 3).

We benchmark five causal post-processing methods:

- **M1 (Baseline Raw Thresholding):** Memoryless frame thresholding:

$$y_{\text{raw}}[n] = \mathbb{I}(s[n] \geq \tau_{\text{raw}}^*). \quad (6)$$

[TODO: Full Validation Threshold Operating Curves for M1–M5]

Precision-Recall and F_1 -vs- τ curves across $\tau \in [0.05, 0.95]$ ($\Delta\tau = 0.02$, 46 candidate thresholds), retained rather than collapsed to a single point at τ_m^* .

Fig. 3. Precision-Recall (or F_1 -vs- τ) operating curves for M1–M5 across the full validation threshold sweep, retained rather than collapsed to a single point at τ_m^* .

Lacking temporal memory, it is vulnerable to transient distractor spikes.

- **M2 (Five-Frame Moving Average):** Causal arithmetic mean:

$$\tilde{s}_{\text{MA}}[n] = \frac{1}{W} \sum_{k=0}^{W-1} s[n-k], \quad (7)$$

asserting alarm $y_{\text{MA}}[n] = \mathbb{I}(\tilde{s}_{\text{MA}}[n] \geq \tau_{\text{MA}}^*)$. A single-frame spike is attenuated by factor $1/W = 0.20$. Moving averages smooth step edges, requiring a lower optimal threshold τ_{MA}^* to maintain recall.

- **M3 (Five-Frame Median Filter):** Order-statistic filter:

$$\tilde{s}_{\text{med}}[n] = \text{median}(s[n], \dots, s[n-4]), \quad (8)$$

asserting alarm $y_{\text{med}}[n] = \mathbb{I}(\tilde{s}_{\text{med}}[n] \geq \tau_{\text{med}}^*)$. It eliminates noise pulses spanning fewer than $\lfloor W/2 \rfloor + 1 = 3$ frames while preserving step edges during target entry.

- **M4 (Five-Frame History Consensus):** Sliding-window M -out-of- N integration ($M = 3, W = 5$):

$$y_{\text{hist}}[n] = \mathbb{I}\left(\sum_{k=0}^{W-1} \mathbb{I}(s[n-k] \geq \tau_{\text{hist}}^*) \geq 3\right). \quad (9)$$

An alarm triggers only if at least 3 of the past 5 frames independently exceed threshold τ_{hist}^* .

- **M5 (Learned Mamba-SSSM):** A pure-PyTorch selective state-space model. Scalar confidence $s[n]$ projects to hidden dimension $d_{\text{model}} = 16$ via SiLU activation: $x_t = \text{SiLU}(W_{\text{in}}s[n] + b_{\text{in}})$. Linear projections yield step size $\Delta_t = \text{softplus}(W_{\Delta}x_t + b_{\Delta})$ and transition matrices $B_t = W_Bx_t$, $C_t = W_Cx_t \in \mathbb{R}^{d_{\text{model}} \times d_{\text{state}}}$ ($d_{\text{state}} = 8$). Diagonal matrix $A = -\exp(A_{\log})$ guarantees stability. Under zero-order hold (ZOH) discretization, recurrence, and alarm assertion:

$$\bar{A}_t = \exp(\Delta_t \cdot A), \quad \bar{B}_t = (\Delta_t \cdot B_t) \odot x_t, \quad (10)$$

$$h_t = \bar{A}_t \odot h_{t-1} + \bar{B}_t, \quad u_t = \sum_{j=1}^{d_{\text{state}}} h_{t,j} \odot C_{t,j}, \quad (11)$$

$$y_{\text{Mamba}}[n] = \mathbb{I}(\sigma(W_{\text{out}}u_t + b_{\text{out}}) \geq \tau_{\text{Mamba}}^*). \quad (12)$$

Mamba-SSSM retains only h_t in memory ($16 \times 8 \times 4$ B = 512 B), ensuring strict $O(1)$ complexity per frame.

Table II contrasts the theoretical complexity and memory footprint of methods M1–M5.

TABLE II
THEORETICAL PER-FRAME COMPLEXITY AND STATE MEMORY
(INDEPENDENT OF HARDWARE)

Method	Time / Frame	State Memory
M1: Raw Thresholding	$O(1)$	0 B (memoryless)
M2: Moving Average	$O(1)$ amortized*	$5 \times 4 \text{ B} = 20 \text{ B}$
M3: Median Filter	$O(W)$ (sorted insert)	$5 \times 4 \text{ B} = 20 \text{ B}$
M4: History Consensus	$O(W)$	5 B (packed booleans)
M5: Mamba-SSSM	$O(d_{\text{model}} \cdot d_{\text{state}})$ (753 params) [†]	$16 \times 8 \times 4 \text{ B} = 512 \text{ B}$

*With a circular buffer, naive resummation is $O(W)$. [†]PyTorch implementation has 753 trainable parameters.

To establish statistical confidence without parametric assumptions over correlated frames, we use non-parametric sequence-level bootstrapping ($B = 1000$ iterations with replacement across the 12 test clips). We report two-sided 95% percentile confidence intervals (2.5th and 97.5th percentiles) in Table III.

Fig. 2 illustrates transient noise rejection versus sustained target confirmation.

D. Downstream Operational Metrics

In aerial search, frame-level false positives incur mission-level costs. Confirmed alarms ($y[n] = 1$) trigger telemetry packets of size S_{pkt} (1.2 kB standard, 45 kB with thumbnail) over satellite or cellular IoT links [3], and require verification loiter maneuvers of duration T_{loiter} (15–30 s) consuming hovering power $P_{\text{hover}} \approx 280 \text{ W}$. Across K_{FP} false alarms, wasted bandwidth and expended battery energy are modeled as:

$$\Omega_{\text{wasted}} = K_{\text{FP}} \cdot S_{\text{pkt}} \quad [\text{bytes}], \quad (13)$$

$$E_{\text{loiter}} = P_{\text{hover}} \cdot T_{\text{loiter}} \quad [\text{Joules}]. \quad (14)$$

We evaluate the False Alarm Suppression Ratio (FASR, Eq. (15)):

$$\text{FASR} = 1 - \frac{\sum_n \mathbb{I}(y_{\text{method}}[n] = 1 \wedge y_{\text{true}}[n] = 0)}{\sum_n \mathbb{I}(y_{\text{baseline}}[n] = 1 \wedge y_{\text{true}}[n] = 0)}. \quad (15)$$

IV. EXPERIMENTAL SETUP AND DATASET

A. Synthetic Multimodal Dataset Specifications

Wilderness SAR dual-spectrum datasets remain scarce. As summarized in Table V, our controlled 2×2 experimental design balances modality and environment across identical scene categorizations.

We evaluate across two wilderness biomes: (1) **Arid Desert**, characterized by high ambient temperatures, rocky terrain, sparse vegetation, and midday thermal crossover where ground temperatures equilibrate with human skin radiation; and (2) **Temperate Forest**, characterized by dense canopies, dappled shadows, intermittent foliage occlusions, and wildlife distractors. Each environment comprises three scenario types: positive clips with human targets (walking, crawling, partially occluded), hard-negative distractors (sun-heated boulders, moving shadows, animal clutter), and clear-negative wilderness terrain.

Raw aerial video is recorded in 1280×720 format at 24 FPS (10.0 s, 240 frames). To preserve ground sampling distance (GSD) without distortion or blurring small targets, we execute

a $2 \times$ dual-crop extraction: from each 1280×720 frame, two disjoint 640×640 snippets are harvested (left tile $x \in [0, 640]$ and right tile $x \in [640, 1280]$, centered at $y = 40$). This yields 30 distinct 10-second sequences per environment (10 positive, 10 hard-negative, 10 clear-negative). Raw MP4 files are remuxed with `-movflags +faststart` to relocate the `moov` atom for streaming in Label Studio. Keyframe bounding boxes were placed at motion inflection points with linear track interpolation. Negative frames are stored as empty 0-byte label files to penalize background false alarms.

B. Sequence Partitioning and Detector Training

Partitioning is conducted strictly at the sequence level to prevent temporal leakage. Clips within each biome and scenario are divided into a 60/20/20 split: 60% training (6 clips/scenario), 20% validation (2 clips/scenario), and 20% testing (2 clips/scenario). Clip frames remain in exact sequential order ($0 \rightarrow 239$) to support sequential modeling. Table VI summarizes the training parameters.

V. RESULTS AND DISCUSSION

A. Upstream Detection Across Modalities and Environments

We evaluate the upstream frame-level detection performance of the two YOLO11n models on the test split. Table VII summarizes precision, recall, and F1-score across modalities and environments under standard thresholding ($\tau = 0.50$).

Qualitatively, the individual streams exhibit complementary trade-offs: (1) **Visible RGB** provides sharp spatial localization in illuminated desert scenes but suffers dropouts under dense tree canopies and shadows in forest scenes; (2) **Thermal infrared** preserves target visibility through foliage occlusions but incurs false positive spikes on solar-heated terrain during midday desert passes; and (3) **Soft disjunctive fusion** via max-pooling increases target recall by propagating detections from either spectrum forward. However, because max-pooling propagates the highest score, raw fusion also passes transient false alarm spikes from both sensors, necessitating causal temporal filtering.

B. Comparative Evaluation of Causal Post-Processing Methods

The five causal post-processing methods are evaluated on the fused confidence streams across the test split under optimal decision thresholds τ_m^* calibrated on validation sequences $\mathcal{D}_{\text{val}}$ via Eq. (5). Table III reports the comparative benchmark. The post-processing methods exhibit distinct filtering dynamics: (1) **M1 (Baseline Raw)**: Lacking temporal memory, memoryless thresholding allows isolated single-frame distractor spikes to trigger false alarms. (2) **M2 (Moving Average)**: Attenuates single-frame spikes by factor $1/W = 0.20$; calibrated threshold τ_{MA}^* adapts downward to compensate for smoothed target onset edges. (3) **M3 (Median Filter)**: Eliminates noise bursts under 3 frames while preserving sharp step edges as targets enter view. (4) **M4 (History Consensus)**: Requires at least 3 of 5 frames to exceed threshold τ_{hist}^* , enforcing discrete temporal consensus against transient clutter. (5) **M5 (Mamba-SSSM)**:

TABLE III

COMPARATIVE BENCHMARK OF CAUSAL TEMPORAL POST-PROCESSING METHODS UNDER VALIDATION-OPTIMIZED DECISION THRESHOLDS ($W = 5$ FRAMES / 208 MS AT 24 HZ)

Method	Val τ^*	Precision	Recall	F1-Score	F1 (95% CI)	Pos. Recall (%)	Hard-Neg FP	Clear-Neg FP	FASR (%)	Latency / Frame
M1: Baseline Raw Thresholding	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M2: Five-Frame Moving Average	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M3: Five-Frame Median Filter	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M4: Five-Frame History Consensus	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M5: Learned Mamba-SSSM	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

TABLE IV

ENVIRONMENT-STRATIFIED FALSE-ALARM SUPPRESSION AND PROJECTED OPERATIONAL SAVINGS

Method	Environment	FASR (%)	FP Avoided vs. M1	Bandwidth Saved (kB)	Battery Saved (kJ)	Est. Added Flight Time (min)
M2: Moving Average	Arid Desert	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M2: Moving Average	Temperate Forest	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M3: Median Filter	Arid Desert	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M3: Median Filter	Temperate Forest	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M4: History Consensus	Arid Desert	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M4: History Consensus	Temperate Forest	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M5: Mamba-SSSM	Arid Desert	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
M5: Mamba-SSSM	Temperate Forest	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

TABLE V

CONTROLLED 2×2 EXPERIMENTAL DESIGN

	Arid Desert	Temperate Forest
RGB (YOLO11n-RGB)	30 clips (10/10/10)	30 clips (10/10/10)
Thermal (YOLO11n-Thermal)	30 clips (10/10/10)	30 clips (10/10/10)

10/10/10 = positive / hard-negative distractor / clear-negative clips.

TABLE VI

DATASET PARTITIONING AND DETECTOR TRAINING PARAMETERS

Parameter / Specification	Configuration Value
Environments & Modalities	Arid Desert, Temperate Forest; Visible RGB, TIR
Raw Video & Timebase	24 FPS (1280 × 720, 10.0 s), $T_s \approx 41.7$ ms (240 f/clip)
Dual-Crop Harvesting	2× square snippets (640 × 640, left & right tiles, $y = 40$)
Sequence Split (Disjoint)	30 clips/env (10 pos/10 hard-neg/10 clear-neg); 60/20/20 split
Upstream Detectors	Two YOLO11n (2.6M params, 6.5 GFLOPs at 640 × 640)
Training Configuration	Batch 16, 100 Epochs, FP32 (amp=False)
Optimizer & Schedule	SGD (momentum 0.937, decay 0.0005), lr ₀ = 0.01 cosine
Sliding Window (W)	5 frames (208 ms causal latency bound)

TABLE VII

FRAME-LEVEL UPSTREAM DETECTION PERFORMANCE ACROSS MODALITIES AND ENVIRONMENTS (TEST SPLIT, $\tau = 0.50$)

Configuration / Condition	Precision	Recall	F1-Score
YOLO11n-RGB (Arid Desert)	[TBD]	[TBD]	[TBD]
YOLO11n-RGB (Temperate Forest)	[TBD]	[TBD]	[TBD]
YOLO11n-Thermal (Arid Desert)	[TBD]	[TBD]	[TBD]
YOLO11n-Thermal (Temperate Forest)	[TBD]	[TBD]	[TBD]
Late Fusion Gate (Desert)	[TBD]	[TBD]	[TBD]
Late Fusion Gate (Forest)	[TBD]	[TBD]	[TBD]

Selectively dissipates isolated spikes through state decay while integrating persistent target motion into confirmed alerts with sub-millisecond per-frame latency.

C. Environmental Robustness and Downstream Operational Impact

Across environmental conditions, distinct operational dynamics emerge: in the **arid desert**, sun-heated boulders and

sand induce thermal crossover spikes that temporal filtering attenuates; in the **temperate forest**, canopy occlusions cause 1-to-2 frame RGB dropouts that soft disjunctive fusion and temporal state memory bridge to sustain target tracks.

Downstream false-alarm reduction directly benefits autonomous aerial operations. On bandwidth-limited IoT links (satellite links at 2.4–10 kbps), suppressing spurious alarms prevents telemetry queue congestion and packet loss [3]. Furthermore, because each alarm mandates a 15–30 s UAV loiter maneuver consuming $P_{\text{hover}} \approx 280$ W, eliminating false alarms conserves tens of kilojoules per sortie, extending flight endurance and search coverage.

Table IV models these savings relative to baseline M1. For each method m and biome $e \in \{\text{Desert}, \text{Forest}\}$, false alarms avoided is $\Delta\text{FP}_m(e) = \text{FP}_{\text{baseline}}(e) - \text{FP}_m(e)$. Via Eq. (13), uplink bandwidth saved is $\Delta\Omega_m(e) = \Delta\text{FP}_m(e) \cdot S_{\text{pkt}}$ ($S_{\text{pkt}} = 1.2$ kB standard, 45 kB thumbnail). Via Eq. (14), battery energy conserved is $\Delta E_m(e) = \Delta\text{FP}_m(e) \cdot P_{\text{hover}} \cdot T_{\text{loiter}}$ ($P_{\text{hover}} \approx 280$ W, $T_{\text{loiter}} \in [15, 30]$ s), yielding added search endurance $\Delta t_{\text{flight}} = \Delta E_m(e) / P_{\text{hover}} = \Delta\text{FP}_m(e) \cdot T_{\text{loiter}}$.

D. Limitations

While providing systematic insights into causal temporal post-processing for aerial SAR, several constraints bound the present findings: (1) **Sample Size**: Evaluated across 12 held-out test clips (2 per scenario per biome, 2,880 frames at 24 Hz); sequence partitioning prevents leakage, though larger flight campaigns are needed for wider generalization. (2) **Biome Diversity**: Constrained to arid desert and temperate forest; maritime, alpine snowpack, wetland, and urban-wildland interface environments introduce distinct scattering not modeled here. (3) **Flight Dynamics**: Sequences reflect nominal flight (30–50 m AGL at 8–15 m/s); turbulent sorties or hover-scans may alter target sampling distance and jitter. (4) **Weather Phenomena**: Clips reflect nominal daytime conditions; dense fog, precipitation, and blowing dust remain

unmodeled. (5) **Single Target:** Formulated for individual person search without multi-target disambiguation or dense group tracking. (6) **Hardware-in-the-Loop:** Latencies and memory bounds are benchmarked in software; physical deployment aboard an NVIDIA Jetson companion computer interfacing with a Pixhawk flight controller remains future validation.

VI. CONCLUSION

This paper investigated lightweight multimodal person detection with causal temporal post-processing for UAV wilderness search and rescue across arid desert and temperate forest biomes using visible RGB and thermal infrared sensing. Two separate YOLO11n baselines (2.6M parameters, 6.5 GFLOPs at 640×640 , batch 16, FP32) were integrated via soft disjunctive late fusion. Downstream, five causal post-processing methods operating across a five-frame (208 ms at 24 Hz) sliding window were benchmarked under validation-calibrated decision thresholds. Causal temporal filtering effectively suppresses transient environmental false alarms while maintaining rapid target confirmation, conserving telemetry bandwidth and onboard battery reserves. Future work will address hardware-in-the-loop testing on NVIDIA Jetson companion computers aboard physical UAVs, multi-target tracking, and adverse weather adaptation.

DATA AND CODE AVAILABILITY

To ensure reproducibility in wilderness SAR, the MMSAR simulation framework, dual-crop snippet harvesting scripts, and pure-PyTorch causal post-processing implementations are archived at [PLACEHOLDER: code repository URL, or “available upon request from the corresponding author”]. Synthetic multimodal video clips and bounding-box manifests are accessible at [PLACEHOLDER: dataset repository URL or DOI]. Pre-trained YOLO11n-RGB and YOLO11n-Thermal model weights are archived under [PLACEHOLDER: model weight repository URL].

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